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BSE-7B

Date:
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PPIT

Case Study: GreenTech Solar Expansion Project

a) Annual depreciation expense = $\frac{\text{cost} - \text{salvage value}}{\text{life}}$

$= \frac{4,000,000 - 500,000}{5} = \$700,000$ per year

b) Annual Net Cash inflows after tax

1. EBT = Sales - Operating costs - depreciation

Earnings Before Tax = 3,000,000 - 1,500,000 - 700,000

= 800,000

2. Calculate tax = 25% × 800,000

= 200,000

3. Net income (Profit after tax) / Earnings After Tax (EAT)

= 800,000 - 200,000 = 600,000

4. Add back Non-Cash depreciation to get net cash inflow

Annual Net cash inflow = net income + depreciation
(Years 1 - 4)
= 600,000 + 700,000 = $1,300,000

~~Annual N~~

In the final year, add salvage value of 500,000

Year 5 total = $1,800,000

c) Payback Period = $\frac{\text{Initial Investment}}{\text{Annual Cash Inflow}} = \frac{4,000,000}{1,300,000}$

= 3.08 years (approx 3 years & 1 month)

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d) Net Present Value (NPV)

- We discount annual cash inflows at 10%.

1. PV of Cash Inflows

Year	Cash Flow	PV factor (10%)	PV of cash flow
1	1,300,000	0.909	1,181,700
2	1,300,000	0.826	1,073,800
3	1,300,000	0.751	976,300
4	1,300,000	0.683	887,900
5	1,800,000	0.621	1,117,800

Total PV of inflows = 5,237,500

2. Subtract Initial Investment

NPV = 5,237,500 - 4,000,000 = **$1,237,500**
(positive)

- Since NPV is positive and the payback period is only 3.08 years (less than 5 years project life), the investment is financially viable & should be accepted